

Auteur: Tala Mohanna
Studierichting: Informatica
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$$\begin{aligned} & \int \cos x \cos 2x dx \\ &= \int \frac{1}{2} (\cos(x - 2x) + \cos(x + 2x)) dx \\ &= \frac{1}{2} \int (\cos(-x) + \cos 3x) dx \\ &= \frac{1}{2} \int (\cos x + \cos 3x) dx \\ &= \frac{1}{2} \left(\int \cos x dx + \int \cos 3x dx \right) \\ &= \frac{1}{2} \left(\sin x + \frac{1}{3} \sin 3x \right) + c \\ &= \frac{1}{2} \sin x + \frac{1}{6} \sin 3x + c \end{aligned}$$

References